Building Agentic RAG Systems with Elasticsearch

Hello I am Dishant!

- Work in the field of AI/ML, focusing on Generative AI and Conversational AI
- Enjoy building chatbots and AI tools that solve everyday problems
- Comfortable with cloud and DevOps tools (Docker, Kubernetes, AWS)
- Often share knowledge through talks and workshops with students & communities
- Curious learner who loves simplifying tech concepts for everyone

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01

Why Elasticsearch?

"You don't need 5 tools to build an AI app.
You need one."

Why Elasticsearch

- Pinecone: cloud-only, expensive at scale, no hybrid search
- ChromaDB: dev-only, not production ready, no observability
- Milvus: complex ops, separate infra, no native LLM tooling
- All of them: you still need 3–4 more tools to build a real app
- With Elastic: one Docker compose file and you're production-ready

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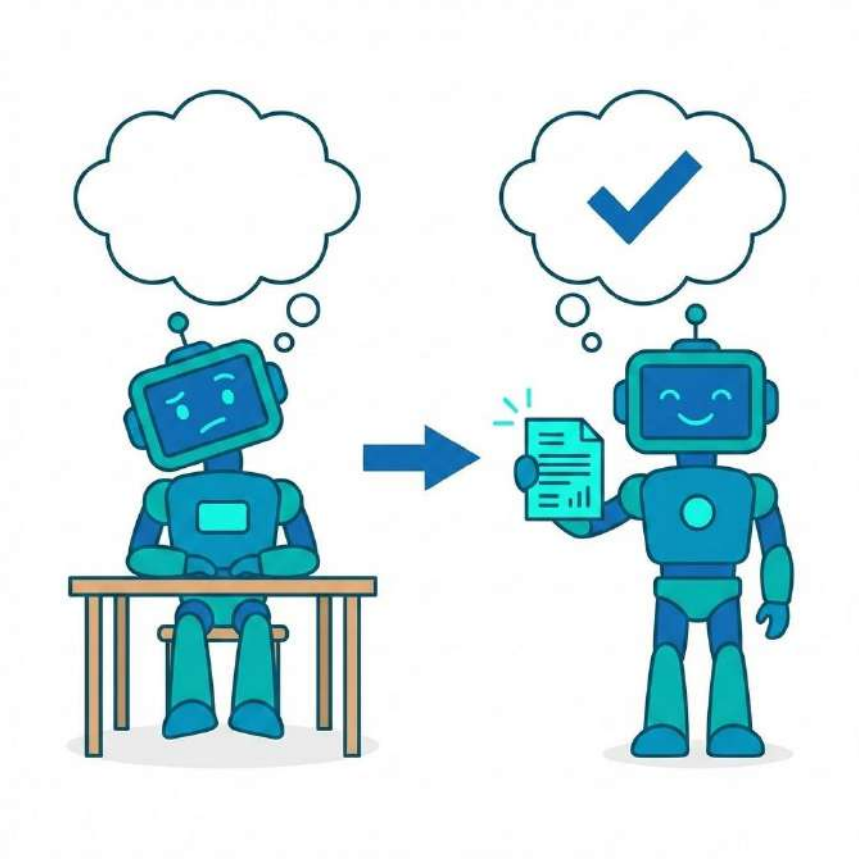
What is RAG?

What is RAG?

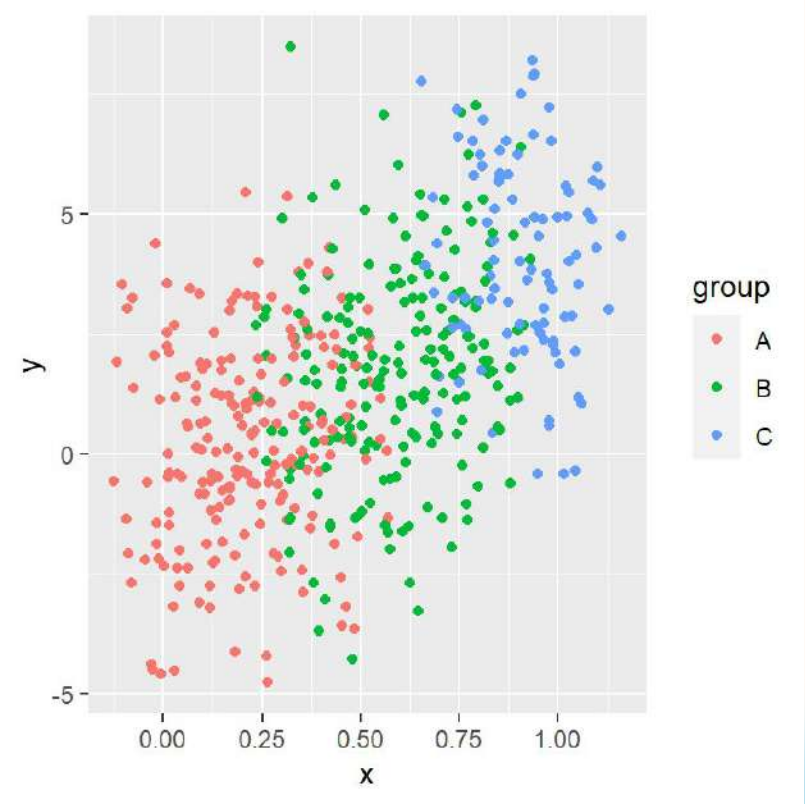


- LLMs are frozen in time — they don't know your data
- RAG = give the LLM a "cheat sheet" before it answers
- The cheat sheet = your documents, retrieved semantically
- Without RAG: LLM guesses → hallucination
- With RAG: LLM reads → grounded answer
- "It's not magic. It's search + generation combined."

What is RAG?

- RAG = Retrieval-Augmented Generation
- LLMs hallucinate without grounding → RAG fixes this
- Pipeline: Documents → Embed → Index → Retrieve → Generate
- Your knowledge base becomes the LLM's context window
- Elasticsearch replaces ChromaDB / Milvus as the vector store

What is Embedding



- Words/sentences converted to numbers (vectors)
- Similar meaning = similar numbers = close in space
- "King" and "Queen" are closer than "King" and "Car"
- 384 or 768 or 1536 numbers per sentence
- Elasticsearch stores and searches these numbers at scale
- The embedding model is the translator between language and math

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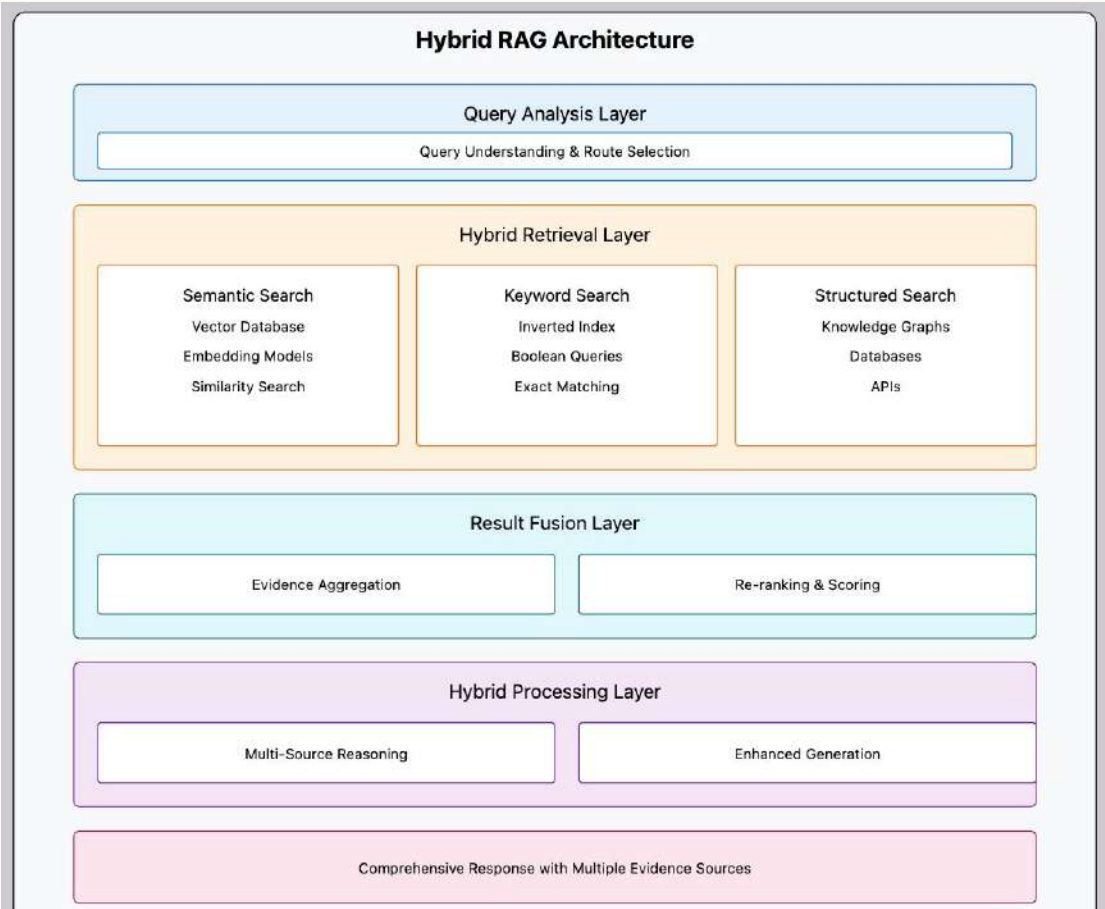
Hybrid RAG

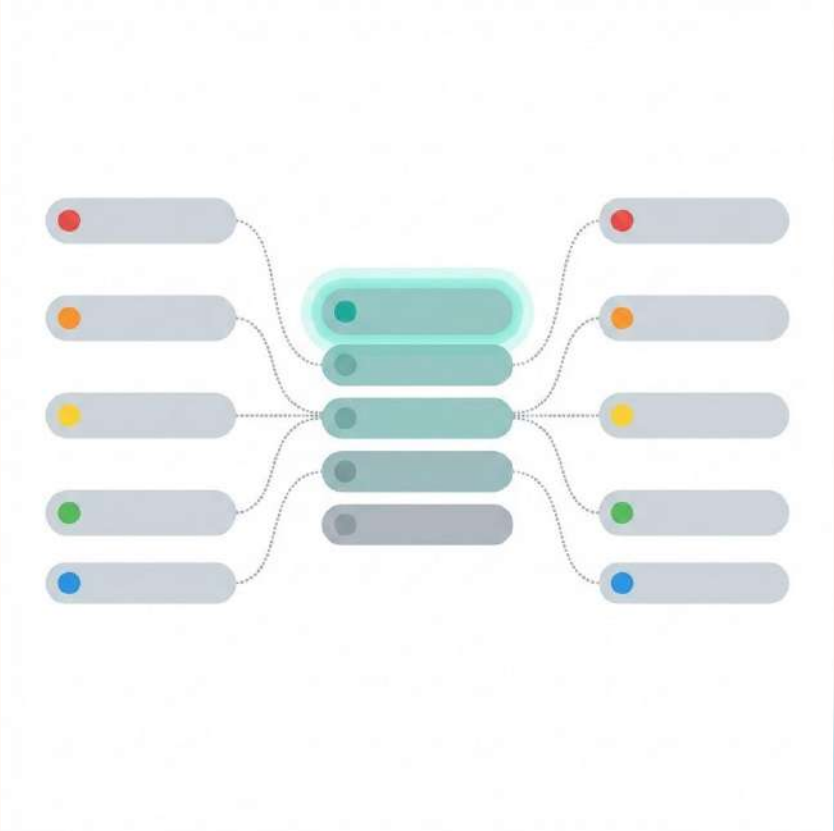
How does it work?



Hybrid search

- Query: "What is the RRF formula in Elasticsearch 8.9?"
- Pure vector search: finds semantically similar docs about ranking — misses the exact version
- Pure keyword search: finds "8.9" and "RRF" — misses the conceptual context
- Hybrid: finds the exact doc about RRF in 8.9 with full context
- This is why production RAG needs hybrid, not just KNN

RRF: How the Merging Works



- BM25 gives ranked list A (by keyword relevance)
- kNN gives ranked list B (by vector similarity)
- RRF formula: score = Σ 1 / (60 + rank)
- A doc ranked #1 in both lists scores highest
- No score normalization needed — ranks are universal
- Result: the best of both search paradigms in one list

AG Quality: What Makes It Good or Bad

Factor	Bad RAG	Good RAG
Chunk size	Too large (loses precision)	256–512 tokens
Search type	Keyword only	Hybrid (BM25 + kNN)
Re-ranking	None	RRF or cross-encoder
Embedding model	Generic	Domain-specific
Context window	Dump everything	Top 3–5 relevant chunks
Observability	None	Kibana traces every query

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Kibana

Kibana

- Kibana acts as the analytics and monitoring interface for the Elastic stack.
- Visualize indexed data in real time
- Monitor search queries and response latency
- Create dashboards for logs, metrics, and traces
- Inspect vector search results and relevance
- Debug ingestion pipelines and document flows

elastic

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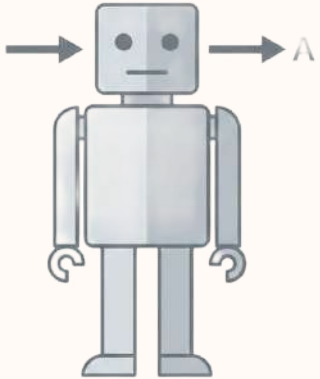
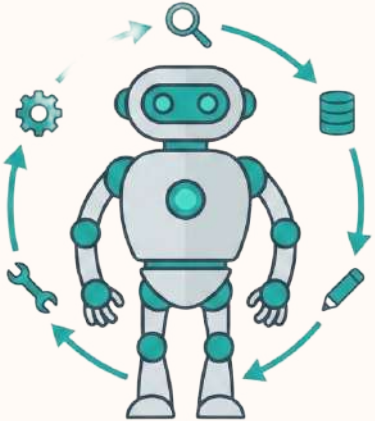
Agentic AI

From RAG to Agentic: What's the Difference?





- RAG: user asks → retrieve → LLM answers → done
- Agentic: user asks → LLM thinks → decides what to retrieve → retrieves → thinks again → answers
- The LLM is no longer a responder. It's a decision maker
- It can loop, use tools, call APIs, query multiple indexes
- Elastic Agent is the orchestration layer that makes this possible

My Agentic AI Stack

- 100% local — no OpenAI, no cloud APIs
- Ollama runs LLaMA / Mistral / Gemma on your laptop
- Elastic runs in Docker
- Total cloud cost: ₹0

How the Agent Decides What To Do

- Agent receives the user query
- Checks: is this a factual lookup or a reasoning task?
- If factual → single ES retrieval → LLM answer
- If complex → multiple retrievals → synthesize → answer
- If uncertain → retrieves more context → re-evaluates
- This loop is what makes it "agentic" vs basic RAG
- "The LLM is the brain. Elasticsearch is the memory."

Why Local LLM + Elastic is Powerful

- Ollama runs Qwen3.5/LLaMA3/Gemma4 locally
- No data leaves your machine — 100% private
- Elastic stores and retrieves your private documents
- Together: a fully private, offline, intelligent RAG agent
- Use cases: internal docs, medical records, legal files, code repos
- Enterprise-grade privacy with zero cloud dependency

What My Agent Can Do

- Answer questions from indexed PDF documents
- Cite the exact source chunk it retrieved
- Handle follow-up questions with context memory
- Run entirely offline on a MacBook / Windows machine
- Observed in real time via Kibana dashboards
- Swap the LLM model in one line (Qwen3.5 → Gemma4)

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Agent Builder

You Don't Have to Code This

- Everything I built manually → Elastic Agent Builder does visually
- No Python, no embeddings code, no Docker compose knowledge needed
- Point-and-click interface inside Kibana
- Connect your data source → choose your LLM → deploy
- Elastic handles chunking, embedding, indexing, retrieval automatically
- "What took me 3 days of setup — takes 20 minutes in Agent Builder"

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Walkthrough

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Ask! Ask!

Lets Connect!

Thanks!

Credits: All icons and images were generated downloaded from AI and slidesgo